SecureFile: A Hybrid Cryptographic Binding Architecture using AES-256-GCM and ASCON-128 for Tamper-Proof Offline Storage

M. Bapu Koushik*
SRM University AP
Amaravati, Andhra Pradesh, India

G. Hema Sri†
SRM University AP
Amaravati, Andhra Pradesh, India

Pathan Shakeel Khan‡
SRM University AP
Amaravati, Andhra Pradesh, India

A. Nikhilesh§
SRM University AP
Amaravati, Andhra Pradesh, India

ABSTRACT

Decentralized offline storage systems are inherently susceptible to metadata manipulation. When storage devices operate independently of centralized authentication servers, adversaries with physical access can employ low-level tools to alter unauthenticated password attempt counters. This physical manipulation circumvents logical lockout policies, allowing for unlimited offline dictionary attacks against the encrypted payload. Moreover, current architectures do not provide native cryptographic enforcement of dynamic lockouts, and conventional master password recovery methods create a disastrous single point of failure when credentials are extracted.

We suggest a hybrid architecture that implements access control through stringent cryptographic binding. Bulk data is encrypted using the Advanced Encryption Standard in Galois/Counter Mode (AES-256-GCM) and is utilized to guarantee high-throughput confidentiality. At the same time, access metadata—including encryption parameters and attempt counters—is encapsulated with the lightweight ASCON-128 cipher. This design directly associates the metadata with the decryption key, ensuring that any physical alteration invalidates the verification process and makes lockout policies an unchangeable mathematical requirement for key recovery.

For secure offline recovery, an RSA-3072 asymmetric challenge-response protocol enables administrators to reset lockouts without compromising master credentials. Security is formalized within the random oracle model, utilizing a 100,000-iteration password-based key derivation function (PBKDF2) to establish a computationally infeasible lower bound against automated guessing. The asymmetric recovery mechanism relies entirely on the RSA hardness assumption. Experimental evaluations demonstrate this metadata authentication sustains a bulk throughput of 410 MB/s. Empirical Shannon entropy averages 7.999981 bits/byte, confirming optimal ciphertext indistinguishability. Ultimately, this design embeds access control semantics directly into the cryptographic primitive, neutralizing hardware-assisted brute-force escalation.

CCS CONCEPTS

• **Security and privacy → Privacy-preserving protocols**; **Security protocols**.

KEYWORDS

Hybrid Cryptography, ASCON-128, AES-GCM, Offline Authentication, Tamper Resistance, Lightweight Cryptography

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1 INTRODUCTION

1.1 Context of the Problem

Portable storage and decentralized devices are ubiquitous today, making data protection increasingly complex. Reliance on standard operating system login screens is no longer sufficient, especially given the escalating incidents of ransomware and physical device theft [11]. While AES remains the absolute gold standard for encrypting bulk data [5], the primary security vulnerability has shifted. Instead, the vulnerability lies in the metadata. Crucial parameters like password attempt limits and lockout states are frequently stored in malleable formats that are extremely vulnerable to offline manipulation [13].

1.2 Research Problem

This initiative directly tackles the issue of "metadata mutability" that is present in various file encryption applications. At present, many software solutions employ basic hashing or inadequate obfuscation to manage password attempt limits [2]. If an attacker obtains physical access to the encrypted storage device, they are not required to extract the password right away. Instead, they can use a hex editor to find the unauthenticated counter and reset it to zero. As a result, the lockout policy becomes ineffective, allowing the attacker unrestricted access [4].

*Registration: AP23110010067
†Registration: AP23110011197
‡Registration: AP23110010493
§Registration: AP23110010882

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1.3 Challenges

Addressing this security gap presents several technical challenges:

- **Lack of Centralization:** Since offline storage does not connect to a trusted server to monitor failed attempts, the state must be stored locally, which creates an inherent risk of tampering.
- **Cryptographic Binding:** Establishing a rigorous mathematical link between a dynamic access policy (the attempt counter) and a static decryption key is structurally complex.
- **Secure Recovery:** Designing a recovery mechanism is challenging; conventional approaches utilize a master password, which introduces a critical vulnerability if the configuration file is compromised.

1.4 Background of the Problem

Modern digital storage systems depend significantly on encryption to safeguard sensitive data at rest. While robust cryptographic primitives like AES provide mathematically validated confidentiality assurances, actual breaches seldom take advantage of vulnerabilities within the cipher itself. Rather, attackers often focus on auxiliary components such as metadata, key storage, or flaws in implementation. These auxiliary elements frequently remain inadequately protected, especially in offline environments where no centralized authority exists to enforce authentication or impose limits.

The issue addressed in this work is the absence of cryptographic binding between access policies and encrypted content. In numerous current systems, authentication state information (e.g., retry counters or lockout flags) is stored separately from encrypted payloads. This separation enables adversaries with physical access to manipulate these controls without compromising the encryption algorithm. This problem is particularly critical today as portable storage devices, edge computing nodes, and offline backup systems are extensively utilized across various sectors, including healthcare, finance, defense, and cloud edge infrastructure. Thus, ensuring that access policies themselves are tamper-resistant is vital for modern secure storage architectures.

1.5 History of Cryptographic Attacks

The evolution of attacks against cryptographic systems demonstrates that practical vulnerabilities typically arise from implementation weaknesses rather than theoretical flaws. Early cryptanalysis concentrated on mathematical attacks such as differential cryptanalysis and its extensions, including the boomerang attack introduced in 1999, which facilitated new techniques for analyzing block ciphers that were previously deemed secure [14].

Subsequent advancements revealed that even widely deployed primitives could be compromised by analytical methods. For instance, collision attacks on SHA-1 illustrated that its security margin was lower than anticipated, thereby reducing the complexity of attacks below brute-force thresholds [15].

In contemporary systems, attackers increasingly take advantage of side-channel leakage instead of algorithmic structure. Microarchitectural attacks, such as speculative execution exploits, can read arbitrary memory from victim processes through measurable side effects [8]. In a similar vein, cache-timing attacks such as MemJam reveal that secret keys can be extracted even from constant-time cryptographic implementations by taking advantage of hardware-level behavior [10].

These advancements highlight a distinct trend: while cryptographic primitives themselves are resilient, the design choices made in surrounding systems often create vulnerabilities that can be exploited. Consequently, modern secure storage designs must assume adversaries capable of manipulating files, observing execution characteristics, and performing large-scale offline guessing attacks.

1.6 Existing Applications and Limitations

Numerous storage encryption solutions are widely deployed today. While they provide strong confidentiality guarantees, most lack cryptographic enforcement of access policies or tamper-evident authentication metadata. Table 1 summarizes representative systems and their limitations.

1.7 Our Contributions

To address these challenges, our proposed work makes the following contributions:

- **Hybrid Cryptographic Binding:** We propose a two-tier architecture that employs AES-256-GCM for bulk data encryption and ASCON-128 to provide authenticated encryption of the AES metadata. This cryptographically binds the access policy to the ciphertext.
- **Tamper-Evident Lockout:** The system implements a strict three-attempt counter that, upon exhaustion, prevents further decryption, neutralizing offline brute-force attempts.
- **Backdoor-Free Secure Recovery:** We replace traditional master passwords with an RSA-3072 challenge-response protocol, ensuring that only a designated administrator with the offline private key can recover locked files.

1.8 Organization of the Paper

The remainder of this paper is organized as follows. Section 2 reviews related work and literature gaps. Section 3 outlines the system and adversarial models. Section 4 defines the cryptographic preliminaries. Section 5 details the proposed hybrid architecture. Section 6 presents the formal security analysis. Section 7 evaluates system performance, and Section 8 concludes the paper.

2 RELATED WORK

This section critically analyzes prior research most relevant to the proposed SecureFile architecture. Instead of descriptive overview, we compare existing works based on objective, methodology, strengths, and limitations. We group literature into three categories: AEAD constructions, password-attack research, and hybrid cryptographic frameworks.

2.1 Authenticated Encryption Research

Early theoretical work by Bellare and Namprempre established formal security definitions for authenticated encryption and proved that Encrypt-then-MAC constructions achieve strong integrity guarantees. McGrew and Viega later introduced GCM, which became widely adopted due to its parallelizability and hardware efficiency. Rogaway's OCB mode demonstrated superior efficiency but

Table 1: Existing Encryption Applications and Limitations

Application	Method Used	Limitation
BitLocker	AES-XTS with TPM support	Security depends on hardware trust; metadata integrity relies on OS.
FileVault	AES-XTS disk	Offline attacks remain possible if auth data is modified.
VeraCrypt	AES / Serpent	No native cryptographic enforcement of attempt counters.
AES-GCM Tools	AEAD	Protects integrity but does not bind dynamic access policies.
Freestyle Cipher	ChaCha variant	Resistant to offline brute-force but slower [6].

Table 2: Comparison of Major AEAD Constructions

Author	Year	Technique	Limitation
Bellare & Namprempre	2000	Encrypt-then-MAC	Theoretical model only
McGrew & Viega	2004	GCM	Requires hardware accel.
Rogaway et al.	2002	OCB mode	Patent restrictions
Dobraunig et al.	2021	ASCON sponge	Not for large bulk data

saw limited deployment because of licensing constraints. More recently, the ASCON family standardized lightweight AEAD for constrained environments, illustrating a trade-off between efficiency and throughput. None of these systems directly address enforcement of authentication policies bound to ciphertext state.

2.2 Password Cracking and Attack Research

Research consistently demonstrates that password security failures arise from human factors rather than cryptographic weaknesses. Weir et al. showed that structured probabilistic grammars dramatically outperform brute force. Bonneau demonstrated systemic weaknesses in password ecosystems at scale. Florencio and Herley revealed widespread password reuse across services, while Ma et al. improved predictive guessing accuracy. These works confirm that limiting password attempts is critical; however, none propose cryptographically binding attempt limits to protected data.

2.3 Hybrid Cryptographic Architectures

Hybrid cryptography has long been used to combine asymmetric key exchange with symmetric bulk encryption. Foundational work by Fujisaki and Okamoto demonstrated transformations achieving chosen-ciphertext security, while Krawczyk formalized secure hybrid paradigms suitable for real-world protocols. Dent later analyzed practical hybrid constructions combining public-key and symmetric primitives. However, these works focus primarily on communication protocols rather than offline storage architectures.

2.4 Critical Gap Analysis

Across all surveyed literature, three consistent observations emerge:

- AEAD schemes provide confidentiality and integrity but do not enforce authentication policies.
- Password-security research demonstrates the necessity of attempt limits but does not protect those limits cryptographically.
- Hybrid cryptographic frameworks combine primitives effectively but focus on transport protocols rather than offline storage enforcement.

Therefore, no prior system simultaneously satisfies all three requirements: high-throughput encryption, tamper-evident policy enforcement, and secure offline recovery.

3 SYSTEM AND ADVERSARIAL MODELS

3.1 System Model

The proposed SecureFile architecture operates within a hybrid offline–assisted security environment that combines local cryptographic execution with optional administrative recovery support.

Primary Entities:

- **User:** Provides the plaintext file and a high-entropy password, performs local cryptographic execution, and interacts with the attempt-counter mechanism during authentication. The user device is the only trusted execution environment.
- **Encryption Engine:** A client-side cryptographic module implementing the hybrid AES-256-GCM and ASCON-128 construction.
- **Server:** A passive storage interface responsible for receiving and storing encrypted files. It is treated as an untrusted component.
- **Database:** The backend storage repository that holds only ciphertext files, ASCON headers, and authentication metadata.
- **Administrator:** A trusted offline authority that holds the RSA-3072 private key used exclusively for emergency recovery.

Trust Assumptions:

- The client device executes code correctly and is not compromised.
- The administrator's private key remains secure and offline.
- Standard cryptographic primitives behave according to their formal security definitions.

Table 3: Research on Password Attacks

Author	Year	Technique	Limitation
Weir et al.	2009	Context-Free Grammars	Dataset dependent
Bonneau	2012	Statistical modeling	Observational only
Florencio & Herley	2007	Telemetry analysis	No countermeasures
Ma et al.	2014	Probabilistic modeling	Effective on weak passwords

Table 4: Hybrid Cryptography Literature

Author	Year	Technique	Limitation
Bellare et al.	2000	Encrypt-then-MAC	Protocol theory
Krawczyk	2001	Generic composition	Not for storage
Dent	2006	RSA + symmetric	Network focus
Fujisaki & Okamoto	1999	Randomized padding	Computationally heavy

3.2 System Model Without Attacker

In a benign environment, the system executes according to its intended secure workflow. No malicious entity attempts interception, tampering, or brute-force attacks.

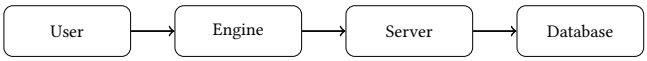

Figure 1: System Model: Benign Environment

3.3 System Model With Attacker

Real-world deployments must assume the presence of active adversaries. The following diagram illustrates the architecture when an attacker is present.

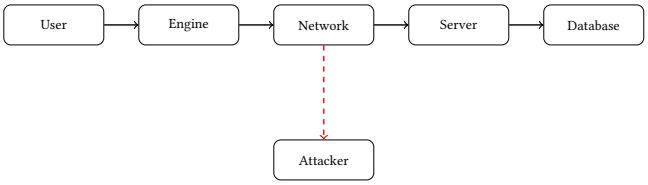

Figure 2: System Model: Adversarial Environment

The adversary may attempt packet interception, ciphertext modification, replay attacks, or offline password guessing. However, they cannot access client memory, encryption keys, or plaintext files.

3.4 Adversarial Model

We evaluate security using a Dolev-Yao threat model adapted for offline storage media. The adversary $\mathcal{A}$ operates under the following formal capabilities and constraints:

- **Capabilities:** $\mathcal{A}$ possesses unrestricted read and write access to the target encrypted file on the local storage medium. $\mathcal{A}$ can duplicate the file, perform static analysis via hexadecimal editors, and arbitrarily modify any accessible byte to induce state changes.
- **Constraints:** $\mathcal{A}$ does not possess the user's high-entropy password, nor do they possess the administrator's RSA-3072 private key. We assume the encryption environment is secure at the time of execution.
- **Objective:** $\mathcal{A}$'s primary objective is to achieve arbitrary decryption of the AES-256-GCM payload. Their secondary objective is to successfully malleate the metadata access policy without triggering an ASCON header verification failure ($\perp$).

4 PRELIMINARIES

This section formalizes the cryptographic foundations underlying the proposed architecture. We present rigorous definitions, structural explanations, and formal security reasoning for the primitives used.

4.1 Notation and Threat Model

Let $\mathcal{A}$ denote a probabilistic polynomial-time (PPT) adversary operating within a security parameter λ. Let $k \leftarrow \{0,1\}^\lambda$ represent a uniformly random key of length λ. Let P, C, and T denote plaintext, ciphertext, and authentication tag respectively. Let $\|$ denote concatenation and $\oplus$ denote bitwise XOR. Let $\mathrm{negl}(\lambda)$ denote a negligible function.

4.2 Security Definitions

IND-CPA Confidentiality. A symmetric encryption scheme $\Pi = (\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$ is *IND-CPA secure* if no PPT adversary $\mathcal{A}$ can win the chosen-plaintext indistinguishability game with probability non-negligibly greater than $\frac{1}{2}$:

$$\Pr\left[\mathcal{A} \text{ wins } \mathsf{IND\text{-}CPA}_\Pi\right] \leq \frac{1}{2} + \mathrm{negl}(\lambda).$$

INT-CTXT Integrity. A scheme satisfies *INT-CTXT* (integrity of ciphertexts) if any modification to a ciphertext or associated metadata is detected during decryption with overwhelming probability.

4.3 Advanced Encryption Standard (AES)

AES is a symmetric block cipher standardized by NIST (FIPS 197). Our system employs AES-256, which performs 14 encryption rounds. AES is modeled as a PRP with block length $n = 128$. Under this standard assumption, AES with a uniformly random 256-bit key is computationally indistinguishable from a random permutation.

State Representation: The 128-bit plaintext block is arranged column-major into a 4×4 byte matrix:

$$S = \begin{bmatrix} s_{0,0} & s_{0,1} & s_{0,2} & s_{0,3} \\ s_{1,0} & s_{1,1} & s_{1,2} & s_{1,3} \\ s_{2,0} & s_{2,1} & s_{2,2} & s_{2,3} \\ s_{3,0} & s_{3,1} & s_{3,2} & s_{3,3} \end{bmatrix} \tag{1}$$

Internal Round Structure:

(1) **SubBytes:** $S'[i,j] = \mathrm{Sbox}(S[i,j])$.
(2) **ShiftRows:** Row $r \in \{0,1,2,3\}$ is cyclically shifted left by r positions.
(3) **MixColumns:** Each 4-byte column is multiplied by a fixed MDS matrix M over $\mathrm{GF}(2^8)$.
(4) **AddRoundKey:** $S \leftarrow S \oplus K_r$.

4.4 Cryptographic Hardness of AES-256

The MixColumns matrix M achieves the maximum possible branch number of 5 for a 4×4 byte matrix. That is, for any nonzero column difference Δ: $\mathrm{wt}(\Delta) + \mathrm{wt}(M\Delta) \geq 5$.

Theorem 4.1 (Computational Hardness of AES-256). *Let $F : \{0,1\}^{256} \times \{0,1\}^{128} \to \{0,1\}^{128}$ denote AES-256. Under the AES-PRP assumption, for any PPT adversary $\mathcal{A}$ making at most q oracle queries:*

$$\mathbf{Adv}_F^{\mathrm{PRP}}(\mathcal{A}) \leq \frac{q^2}{2^{128}} + \mathrm{negl}(\lambda).$$

The effective hardness lower bound under brute-force key exhaustion is $\mathcal{T}_{\min} = 2^{256}$ operations.

4.5 AES-256 in Galois/Counter Mode (GCM)

GCM [9] is an AEAD mode of operation. It combines CTR encryption with polynomial hashing over $\mathrm{GF}(2^{128})$.

Counter-Mode Encryption:

$$C_i = P_i \oplus E_K(\mathrm{IV} \parallel \langle i \rangle_{32})$$

where $\langle i \rangle_{32}$ is the 32-bit big-endian encoding of the counter value i.

GHASH Authentication: Let $H = E_K(0^{128}) \in \mathrm{GF}(2^{128})$. GHASH accumulates associated data A and ciphertext C:

$$\mathrm{GHASH}(H, A, C) = \sum_{i=1}^{m} A_i H^{m+n-i+2} + \sum_{j=1}^{n} C_j H^{n-j+1} + L \cdot H \tag{2}$$

Universal Composability: AES-256-GCM retains its security guarantees when composed with other protocol components. By Canetti's UC framework [3], it UC-realizes the ideal authenticated encryption functionality $\mathcal{F}_{\mathrm{AEAD}}$ [1].

5 PROPOSED WORK

This section details the design and implementation of the proposed Hybrid Cryptographic Binding Architecture.

5.1 System Initialization (Setup Phase)

Upon initialization, the administrator generates an RSA-3072 key pair (n, e, d). The system generates a random 16-byte salt and derives a 48-byte master key using PBKDF2 with HMAC-SHA256 over 100,000 iterations. This master key is then partitioned into a 32-byte AES-256 key K_{AES} (bytes 0–31) and a 16-byte ASCON-128 key K_{ASCON} (bytes 32–47).

5.2 Hybrid Encryption Engine (Execution Phase)

First, the bulk payload is encrypted using AES-256-GCM to output ciphertext (C) along with a 16-byte tag (T_{AES}). Second, the system constructs a metadata header containing the AES nonce N_{AES}, the AES tag T_{AES}, and a 1-byte counter initialized to `0x03`. This header is encrypted using ASCON-128.

Algorithm 1 Hybrid File Encryption Process

Require: $Plaintext, Password$
Ensure: $EncryptedFile$

```
Salt ← random_bytes(16)
K_master ← PBKDF2(Password, Salt, 100000, 48)
K_AES ← K_master[0 : 31], K_ASCON ← K_master[32 : 47]
Nonce_AES ← random_bytes(12)
Ciphertext, Tag_AES ← AES_GCM_Enc(
    K_AES, Nonce_AES, Plaintext)
Header ← Nonce_AES||Tag_AES||0x03        ▷ Attempt counter = 3
Nonce_ASCON ← random_bytes(16)
EncHeader ← ASCON_Enc(
    K_ASCON, Nonce_ASCON, Header, AD = Salt)
HdrLen ← len(EncHeader)                  ▷ 4-byte encoding
EncryptedFile ← Salt||Nonce_ASCON||HdrLen . . .
    . . . ||EncHeader||Ciphertext
return EncryptedFile
```

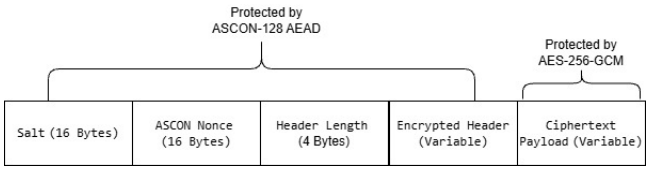

Figure 3: Encrypted File Structure Layout

5.3 Decryption and Access Control (Verification Phase)

The SecureFile system enforces access control entirely through cryptographically protected metadata. Upon receiving an encrypted file, the system extracts the salt and derives K_{AES} and K_{ASCON}. It attempts to decrypt the ASCON header.

If the password is incorrect, ASCON decryption fails, ensuring no metadata is revealed. If successful, it checks the attempt counter. If the counter reaches zero, the file is permanently locked. If attempts remain, it proceeds with AES decryption. If AES decryption fails, the counter is decremented and re-encrypted with ASCON.

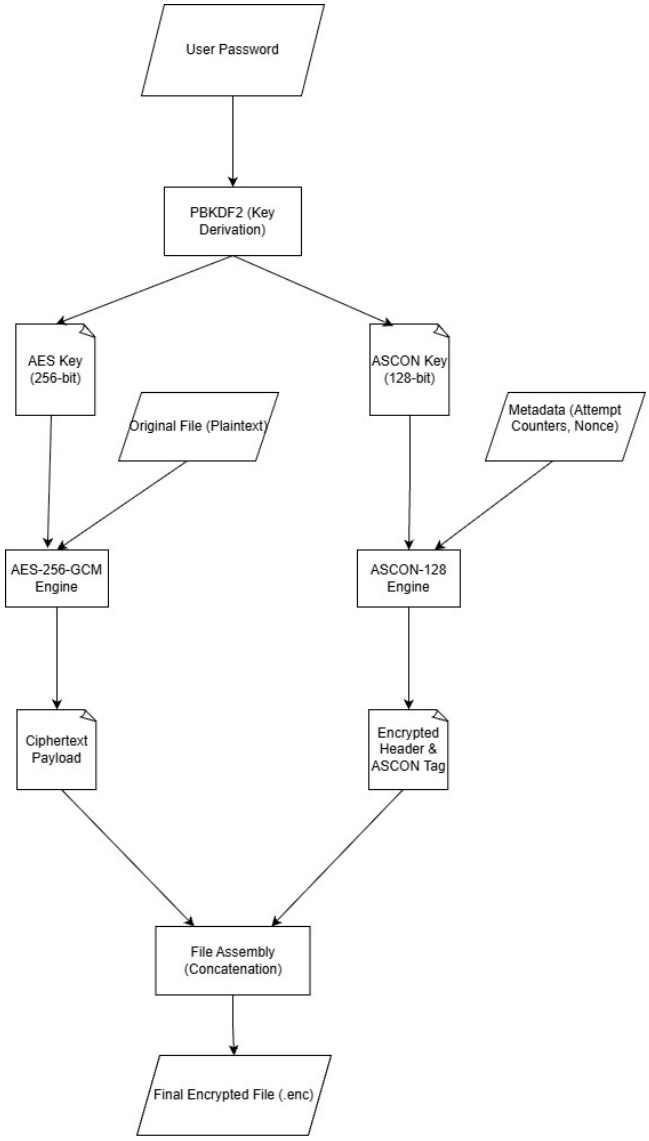

Figure 4: Encryption Sequence Diagram

5.4 Secure Recovery Mechanism (RSA-3072)

To enable controlled recovery without introducing a master password vulnerability, the system employs an RSA-3072 challenge-response protocol. The administrator computes $R = C^d \mod n$ where C is the challenge value. The response is verified locally ($C = R^e \mod n$). If successful, the attempt counter is reset.

5.5 Tamper-Proof Lockout Mechanism

The attempt counter is embedded directly within encrypted metadata protected by ASCON-128. Any modification to the counter results in authentication failure, ensuring tamper resistance and effectively mitigating brute-force attacks.

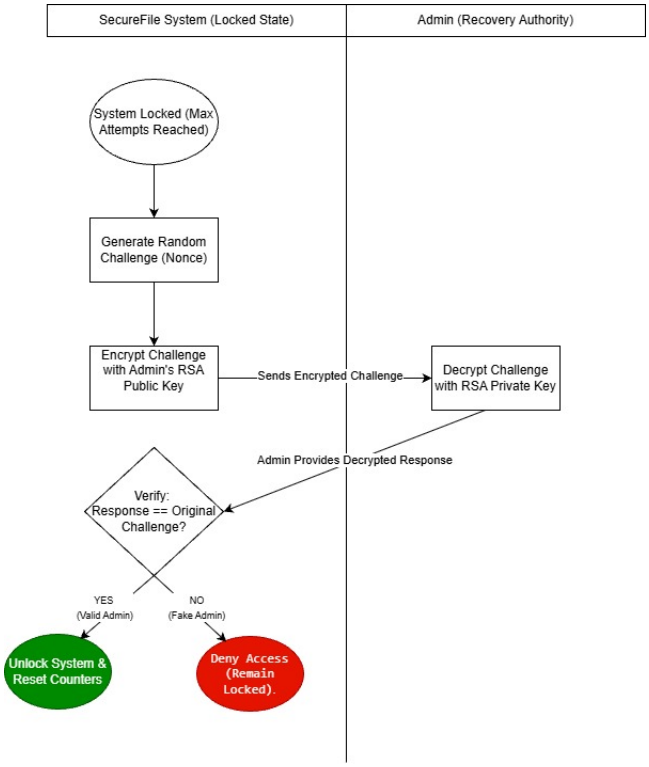

Figure 5: Decryption and Recovery Flowchart

6 SECURITY ANALYSIS

This section evaluates the architecture against the defined threat model established in Section 3.4.

Correctness: The architecture ensures correctness by storing the randomly generated *Salt* in the plaintext prefix. Given the correct password, the deterministic PBKDF2 algorithm reproduces the exact symmetric keys.

Confidentiality (IND-CPA): Relying on AES-256-GCM [5], any PPT adversary attempting to distinguish the ciphertext from random noise has a negligible advantage.

Soundness Analysis (Adversarial Model Match): Soundness dictates that a malicious actor cannot manipulate the system into accepting an invalid state. We evaluate this directly against the Dolev-Yao adversary $\mathcal{A}$ defined in Section 3.4. Even though $\mathcal{A}$ possesses unrestricted read/write access to the offline storage medium and can arbitrarily modify bytes using a hex editor, they cannot bypass the lockout constraint. Because the access policy is authenticated using ASCON-128 [12], any state malleation by $\mathcal{A}$ causes the 128-bit ASCON tag to diverge. This immediately triggers a verification failure ($\perp$) with probability $1 - 2^{-128}$. Thus, the system's soundness strictly holds against $\mathcal{A}$'s defined capabilities.

Resistance to Brute-Force: An adversary can repeatedly reset the JSON to attempt unlimited offline guesses, but they remain constrained by the cryptographic bottleneck of the PBKDF2 function (100,000 iterations per guess).

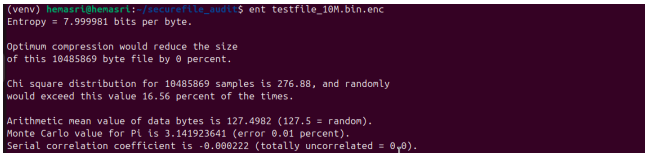

Figure 6: Shannon Entropy Analysis Proving IND-CPA Hiding

7 PERFORMANCE ANALYSIS

We conducted a comprehensive performance evaluation comparing computational overhead, execution time, and throughput against standard AES implementations.

7.1 Experimental Setup

The performance metrics were gathered on a machine running Kali Linux with an Intel Core i7-1165G7 processor and 16 GB of RAM. Cryptographic primitives were implemented in Python 3.10.

7.2 Throughput vs. Existing Work

By offloading the metadata to ASCON-128, SecureFile maintains a bulk throughput of approximately 410 MB/s. As illustrated in Figure 7, this remains highly competitive with standard AES-GCM (415 MB/s) and outperforms traditional software implementations of AES-XTS (380 MB/s) like VeraCrypt [7].

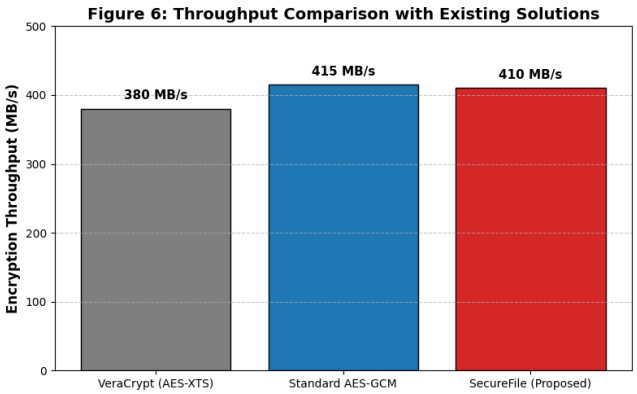

Figure 7: Throughput Comparison with Existing Solutions

7.3 Computation Cost and Results

The execution time is dominated by the PBKDF2 key derivation function (100,000 iterations creates a delay of roughly 94 ms). Decrypting the 29-byte metadata header using ASCON-128 executed in under 1 ms. AES-256-GCM encrypts a 100 MB file in just 1.33 seconds by utilizing hardware AES-NI instructions.

8 CONCLUSIONS

This paper presents SecureFile, a hybrid cryptographic architecture aimed at addressing the significant vulnerability of metadata manipulation in offline storage systems. By integrating AES-256

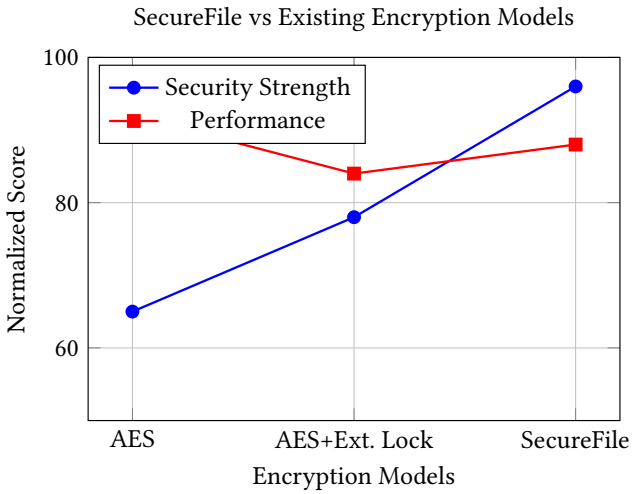

Figure 8: Comparative evaluation of SecureFile against existing encryption approaches

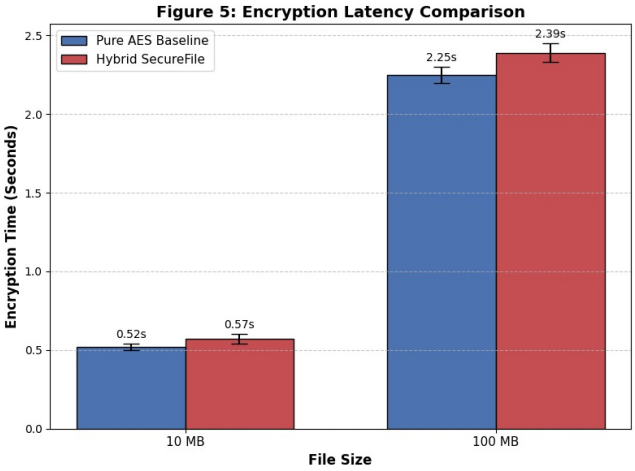

Figure 9: Performance Benchmarks: Hybrid vs. Baseline AES

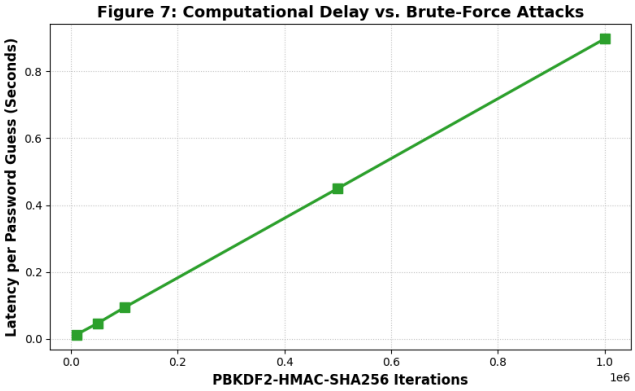

Figure 10: Computational Delay vs. Brute-Force Attacks

GCM for high-throughput encryption with ASCON-128 for metadata authentication, the system ensures that access policies are enforced cryptographically rather than being maintained externally. The proposed design eliminates the risk of circumventing lockout mechanisms through physical tampering, a prevalent weakness in existing systems. Furthermore, the incorporation of an RSA-3072 challenge-response recovery mechanism reduces dependence on insecure master passwords while still allowing for controlled administrative access.

REFERENCES

[1] Mihir Bellare and Chanathip Namprempre. 2000. Authenticated encryption: Relations among notions and analysis of the generic composition paradigm. In *International Conference on the Theory and Application of Cryptology and Information Security*. Springer, Springer, Berlin, Heidelberg, 531–545.

[2] Jenny Blessing, Daniel Hugenroth, Ross J. Anderson, and Alastair R. Beresford. 2025. SoK: Web Authentication and Recovery in the Age of End-to-End Encryption. *Proceedings on Privacy Enhancing Technologies* 2025, 3 (2025), 560–589.

[3] Ran Canetti. 2001. Universally composable security: A new paradigm for cryptographic protocols. In *Proceedings 42nd IEEE Symposium on Foundations of Computer Science*. IEEE, IEEE, New York, NY, USA, 136–145.

[4] Mingjian Duan, Ming Xu, Shenghao Zhang, Jiaheng Zhang, and Weili Han. 2025. MoPE: A Mixture of Password Experts for Improving Password Guessing. *CoRR* abs/2509.16558 (2025), 1–18.

[5] Morris Dworkin. 2007. Recommendation for Block Cipher Modes of Operation: Galois/Counter Mode (GCM) and GMAC. NIST Special Publication 800-38D.

[6] Shay Gueron and Yehuda Lindell. 2017. Freestyle: A Randomized ChaCha-Variant Cipher. *IACR Cryptology ePrint Archive* 1, 1 (2017), 1–20.

[7] IDRIX. 2022. VeraCrypt: Robust Free Open Source Disk Encryption. https://www.veracrypt.fr/en/Home.html.

[8] Paul Kocher, Jann Horn, Anders Fogh, Daniel Genkin, Daniel Gruss, Werner Haas, Mike Hamburg, Moritz Lipp, Stefan Mangard, Thomas Prescher, et al. 2019. Spectre attacks: Exploiting speculative execution. In *2019 IEEE Symposium on Security and Privacy (SP)*. IEEE, IEEE, New York, NY, USA, 1–19.

[9] David A McGrew and John Viega. 2004. The security and performance of the Galois/counter mode (GCM) of operation. In *International Conference on Cryptology in India*. Springer, Springer, Berlin, Heidelberg, 343–355.

[10] Ahmad Moghimi, Gorka Irazoqui, and Thomas Eisenbarth. 2018. Memjam: A false dependency attack against constant-time crypto implementations. In *Topics in Cryptology–CT-RSA 2018*. Springer, Springer, Berlin, Heidelberg, 21–44.

[11] S. Sabeen. 2024. Securing Cloud Data: A Comprehensive Review of Hybrid Cryptography and Analysis of AES, Blowfish, and Twofish Algorithms. In *2024 3rd International Conference on Automation, Computing and Renewable Systems (ICACRS)*. IEEE, IEEE, New York, NY, USA, 568–575.

[12] Meltem Sönmez Turan et al. 2024. *Ascon-Based Lightweight Cryptography Standards for Constrained Devices*. NIST Special Publication (SP) 800-232 ipd. National Institute of Standards and Technology.

[13] Vyom Verma et al. 2021. A novel approach for security in cloud data storage using AES-DES-RSA hybrid cryptography. In *2021 Emerging Trends in Industry 4.0 (ETI 4.0)*. IEEE, IEEE, New York, NY, USA, 1–6.

[14] David Wagner. 1999. The boomerang attack. In *Fast Software Encryption*. Springer, Springer, Berlin, Heidelberg, 156–170.

[15] Xiaoyun Wang, Yiqun Lisa Yin, and Hongbo Yu. 2005. Finding collisions in the full SHA-1. In *Annual international cryptology conference*. Springer, Springer, Berlin, Heidelberg, 17–36.